

Sreenithya Mallavarapu

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EDUCATION

University of Wisconsin Madison

Masters in Biological Systems Engineering

Madison, Wisconsin

Sep 2024 – May 2026

Indian Institute of Technology Jodhpur

Bachelor of Technology in Bioscience and Bioengineering

Rajasthan, India

Dec 2020 – May 2024

RELEVANT COURSEWORK

Biostatistics, Computational and Systems Biology, Biophysics and structural biology, Biochemistry, Introduction to Machine Learning, Deep Learning, Differential Equations, Probability, statistics and stochastic processes

EXPERIENCE

University of Wisconsin Madison

Graduate Student Researcher

October 2024 - Present

Madison, WI

- Investigating the interactions of GABA with vitamins and short-chain fatty acids along the bloodstream and vagal passage, aiming to understand its role in neuronal function.
- Conducting computational screening of molecular interactions, followed by in vitro validation using brain cell lines to assess bioavailability and neuroactive effects.
- Developed expertise in molecular docking, statistical and data analysis for studying neuroactive compounds.

Advanced Centre for Treatment, Research, and Education in Cancer (ACTREC) June -July 2023

Summer Intern

Mumbai, India

- Developed and implemented a machine learning algorithm to predict the pathogenicity of PTEN gene mutations associated with breast cancer.
- Compiled a comprehensive training dataset using computational bioinformatics tools such as PolyPhen-2, SIFT, and Align GVGD, quantifying mutation impact on a 0-1 scale.
- Optimized algorithm performance and validation through Python, Orange Tool, and Microsoft Excel.
- Technical Skills: Python, Machine Learning, Bioinformatics Tools, Data Analysis

Indian Institute of Technology Bombay

Summer Intern

May -July 2022

Mumbai, India

- Conducted expression, purification, and crystallization of Topoisomerase (Topo67 WT) protein.
- Gained hands-on experience with advanced biochemical techniques, including SDS-PAGE, FPLC, Ni-NTA chromatography, and UV spectrophotometry.
- Collaborated on X-ray diffraction studies for protein-ligand binding analysis using computational modeling.
- Technical Skills: Protein Purification, Crystallization, Computational Drug Discovery, Spectrophotometry

PROJECTS

Genetic Architecture Analysis Using LD Score Regression | Graduate Coursework

Feb 2025 – Present

- Performed genetic correlation and heritability estimation using the LDSC package. Analyzed publicly available GWAS summary statistics for traits including IBD, SCZ, EA, and AF.
- Conducted meta-analysis of GWAS datasets and explored cross-trait genetic correlations.
- Utilized R, bash scripts, and LDSC tools for interpreting outputs (heritability estimates, intercepts, rg values).

Deep Learning-Based Quantification of Tunneling Nanotubes (TNTs) | B.Tech Project Aug 2023 – May 2024

- Developed a deep learning model using YOLO on the COCO dataset to quantify TNTs in brain tumor cells from image datasets (400 brain tumour cells).
- Performed image augmentation in RoboFlow to enhance detection accuracy.
- Implemented compactness and circularity as features for scoring for cell spheroids.

TECHNICAL SKILLS

Languages: Python, R, SQL, C

Databases: NCBI, NGS, Swiss-prot, PDB, UniProt

Developer Tools: VS Code, Colab, Pymol, Pyrx, BLAST, FASTA, Microsoft office, Linux

Libraries: pandas, LDSC, NumPy, Matplotlib